Identification of Anions

Exam code: 0620 & 0971

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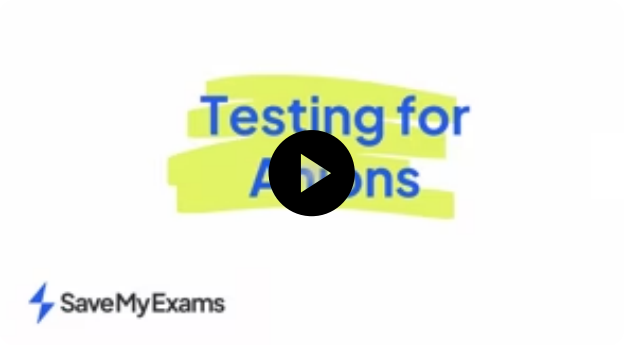
Updated on 18 April 2025



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Identification of anions

- Negatively charged non-metal ions are known as **anions**
- You must be able to **describe** the tests for the following ions:
  - Carbonate ions, CO_3^{2-}
  - Halide ions, Cl^-, Br^-, I^-
  - Nitrate ions, NO_3^-
  - Sulfate ions, SO_4^{2-}
  - Sulfite ions, SO_3^{2-}

Help me understand

Test for carbonate ions

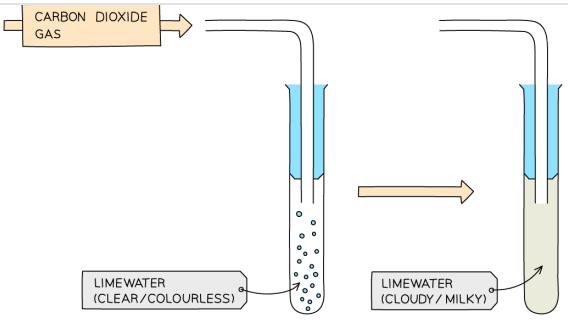
- Carbonate compounds contain the carbonate ion, CO_3^{2-}
- The test for the carbonate ion is:
  - Add **dilute acid**
  - Bubble the **gas** released through limewater
  - Limewater turns cloudy if the carbonate ion is present
- If a carbonate compound is present then fizzing / effervescence should be seen as **CO_2** gas is produced, which forms a white precipitate of calcium carbonate when bubbled through limewater:

$$CO_3^{2-}(aq) + 2H^+(aq) \rightarrow CO_2(g) + H_2O(l)$$

$$CO_2(g) + Ca(OH)_2(aq) \rightarrow CaCO_3(s) + H_2O(l)$$

- The white precipitate turns limewater **cloudy**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



Limewater turns milky in the presence of carbon dixoide caused by the formation of insoluble calcium carbonate

Examiner Tips and Tricks

- If you are asked to describe the test for carbonate ions, make sure that you say:
  - Bubble the gas produced through limewater, which turns cloudy if the carbonate ion is present
- Just saying that limewater turns cloudy is not enough
  - This isn't describing the test, it is stating the result

Test for halide ions

- Halide ions are the negative ions / anions formed by the elements in Group 7
- The test for the halide ions is:
  - Acidify the sample with nitric acid
  - Add **silver nitrate solution,** $AgNO_3$,
  - A silver halide **precipitate** forms if a halide ion is present
    - The precipitate is indicated by the state symbol (s)

- The colour of the silver halide precipitate depends on the halide ion:
  - The chloride ion forms a **white** precipitate of silver chloride

potassium chloride + silver nitrate → potassium nitrate + silver chloride

$$KCl\,(aq) + AgNO_3\,(aq) \rightarrow KNO_3\,(aq) + AgCl\,(s)$$

- The bromide ion forms a **cream** precipitate of silver bromide

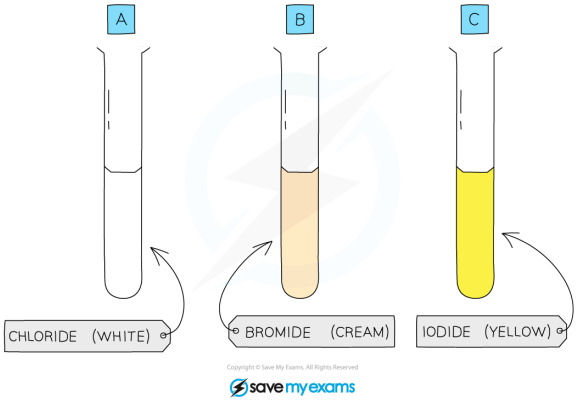
potassium bromide + silver nitrate → potassium nitrate + silver bromide

$$KBr\,(aq) + AgNO_3\,(aq) \rightarrow KNO_3\,(aq) + AgBr\,(s)$$

- The iodide ion forms a **yellow** precipitate of silver iodide

$$KI(aq) + AgNO_3(aq) \rightarrow KNO_3(aq) + AgI(s)$$

Testing for halide ions



Each silver halide produces a precipitate of a different colour

Examiner Tips and Tricks

The acidification step in the halide ion test must be done with nitric acid rather than hydrochloric acid.

HCl contains the chloride ion which would interfere with the results.

Test for nitrate ions

- Nitrate compounds contain the nitrate ion, NO_3^-
- The test for the nitrate ion is
  - Add **aqueous NaOH** and **aluminium foil**
  - **Warm** gently and test the gas released
  - The gas given off is ammonia, NH3
- Ammonia is a gas with a characteristic sharp choking smell that turns damp **red** litmus paper **blue**

Test for sulfate ions

- Sulfate compounds contain the sulfate ion, SO_4^{2-}
- The test for the sulfate ion is:
  - Acidify the sample with **dilute nitric acid**
  - Add a few drops of **barium chloride solution**
  - A **white** precipitate of barium sulfate is formed, if the sulfate ion is present

$$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$$

- The test can also be carried out with barium nitrate solution

A white precipitate of barium sulfate is a positive result for the presence of sulfate ions

Examiner Tips and Tricks

Nitric is added first to remove any **carbonates** which may be present which would also produce a precipitate and interfere with the results.

Test for sulfite ions

- Sulfite compounds contain the sulfite ion, SO_3^{2-}
- The test for the sulfite ion is:
  - Add **dilute acid**
  - Warm the mixture gently
  - Bubble the **gas** released through potassium manganate(VII) solution
  - The potassium manganate(VII) solution changes from purple to colourless if the sulfite ion is present

Examiner Tips and Tricks

For qualitative inorganic analysis, there will be one test for the metal **cation** and another test for the non-metal **anion**.

If you are an extended level student you may be asked to write balanced ionic equations for cation and anions tests, so make sure you know the formulae of all the ions and precipitates formed.

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